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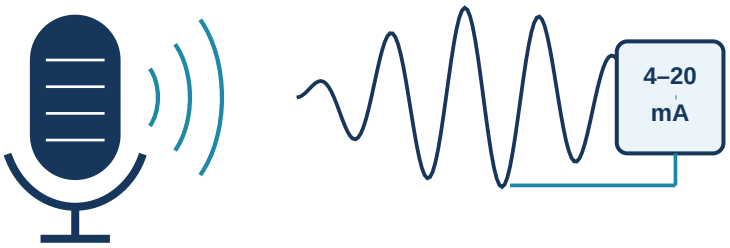
Ahmedabad

Affiliated to Gujarat Technological University (GTU)

PROJECT REPORT ON

Analog Sound Level Measurement

Using Electret Microphone and 4–20 mA Industrial Standard



Subject Name	Industrial Measurement II
Subject Code	BE05000041
College	L. D. College of Engineering, Ahmedabad
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01 Introduction

Sound level is the logarithmic measure of sound pressure, expressed in decibels sound pressure level (dB SPL) relative to 20 µPa. In industrial instrumentation it is monitored for noise-exposure control, machinery condition monitoring and detection of abnormal events.

This project is an analog sound level meter for the 60–90 dB SPL range. An electret microphone signal is amplified, filtered, rectified, averaged and converted into a 4–20 mA current, the standard signal for sending measurements to a PLC or DCS over long cables with good noise immunity.

Scope note: The 4–20 mA output represents 60–90 dB only after calibration against a reference sound source and validation at several levels. Before that, it is only a sound-related amplitude signal. The circuit is not a standards-compliant sound level meter (IEC 61672).

02 Objectives

- To convert sound pressure into an electrical signal with an electret microphone.
- To amplify and band-limit the signal (about 106 Hz – 10.6 kHz) using an AD620 and Sallen-Key filters.
- To rectify and average the signal into a DC level related to sound amplitude.
- To scale this level to 0.4 – 2.0 V and convert it into a 4 – 20 mA current.
- To simulate the circuit in LTspice and build it on a breadboard.
- To identify the calibration needs and design limitations.

03 Components Required

No.	Component	Qty	Purpose
1	Electret condenser microphone (e.g. CZN-15E)	1	Converts sound pressure to a small AC voltage
2	AD620 instrumentation amplifier	1	Front-end gain; set by one resistor (RG)
3	TL084 quad op-amp	3	Two Sallen-Key filter sections and two precision rectifier sections
4	OP07 precision op-amp	2	Gain/offset stage and current-converter error amplifier
5	2N3904 NPN transistor	1	Pass transistor of the voltage-to-current converter
6	100 Ω sense resistor (0.1 %)	1	Sets the relation I = V / 100 Ω
7	Resistors (10 kΩ, 5 kΩ, 1 kΩ, 500 Ω, 2 kΩ, etc.)	As required	Filters, rectifier, gain setting, base resistor
8	Capacitors (0.15 µF, 1.5 nF, 10 µF, 0.1 µF)	As required	Coupling, filtering, averaging, supply bypass
9	Signal diodes (e.g. 1N4148)	2	Precision rectifier
10	Multi-turn trimmer with fixed resistor	1	Fine calibration of the gain stage
11	Reference divider network	1	mV-level offset reference for the 4 mA point
12	LM7812 and LM7912 regulators	1 each	±12 V supply rails
13	24 V DC loop supply and load resistor	1 each	Powers the current loop; load simulates the receiver
14	Breadboard, wires, two digital multimeters	—	Assembly and measurement

Quantities are indicative. Final values are listed in the schematic and may change after calibration.

04 Theory

A pressure-type electret microphone produces a voltage proportional to sound pressure. Because dB is logarithmic and pressure is linear, the measuring chain converts the AC signal into a DC level proportional to its average amplitude. A current loop then transmits this level; the 4 mA live zero keeps the lowest reading above 0 mA.

Stage	Governing relation	Remark
Sound pressure level	Lp = 20 log10(p / 20 µPa)	60 dB = 0.020 Pa; 90 dB = 0.632 Pa
Microphone	V = S × p, S = −44 dBV/Pa ≈ 6.3 mV/Pa	0.126 mV RMS at 60 dB; 3.98 mV at 90 dB (documentation value; confirm with datasheet)
Instrumentation amplifier	G = 1 + 49.4 kΩ / RG	One resistor sets the gain
Sallen-Key filters	fn = 1 / (2πRC)	106 Hz (10 kΩ, 0.15 µF); 10.6 kHz (10 kΩ, 1.5 nF)
Rectifier and averaging	Vdc = 2Vp / π ≈ 0.637 Vp	Average-responding detector, ≈ 0.9 × RMS for a sine wave
Gain and offset	Vcontrol = K (Vdc + Vref)	K = 1.6 V / (Vdc,90 − Vdc,60)
Voltage-to-current	I = Vcontrol / Rsense	0.4 V → 4 mA; 2.0 V → 20 mA with 100 Ω

05 Block Diagram

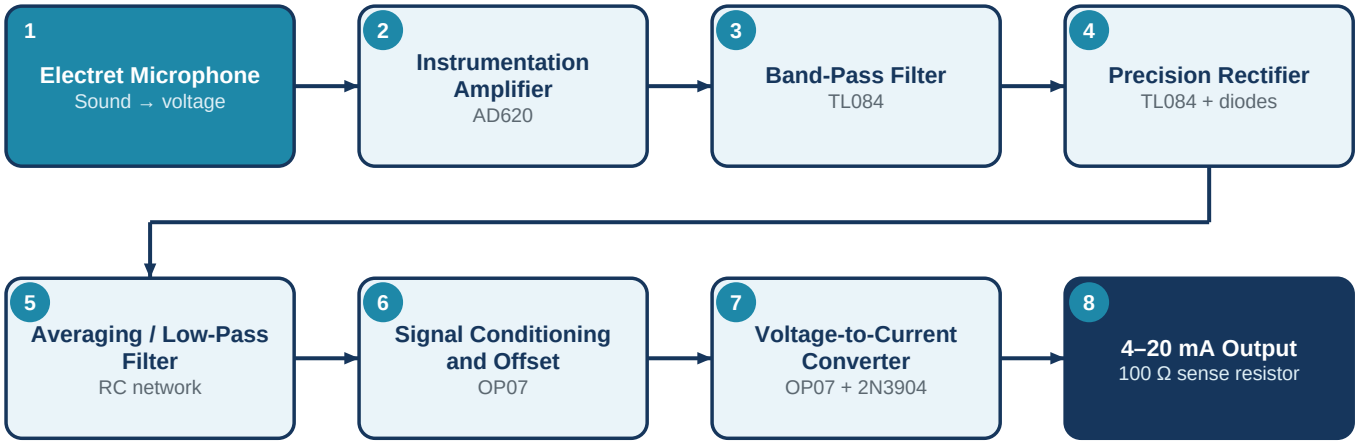


Figure 1: Signal flow from microphone to 4–20 mA output.

06 Flowchart

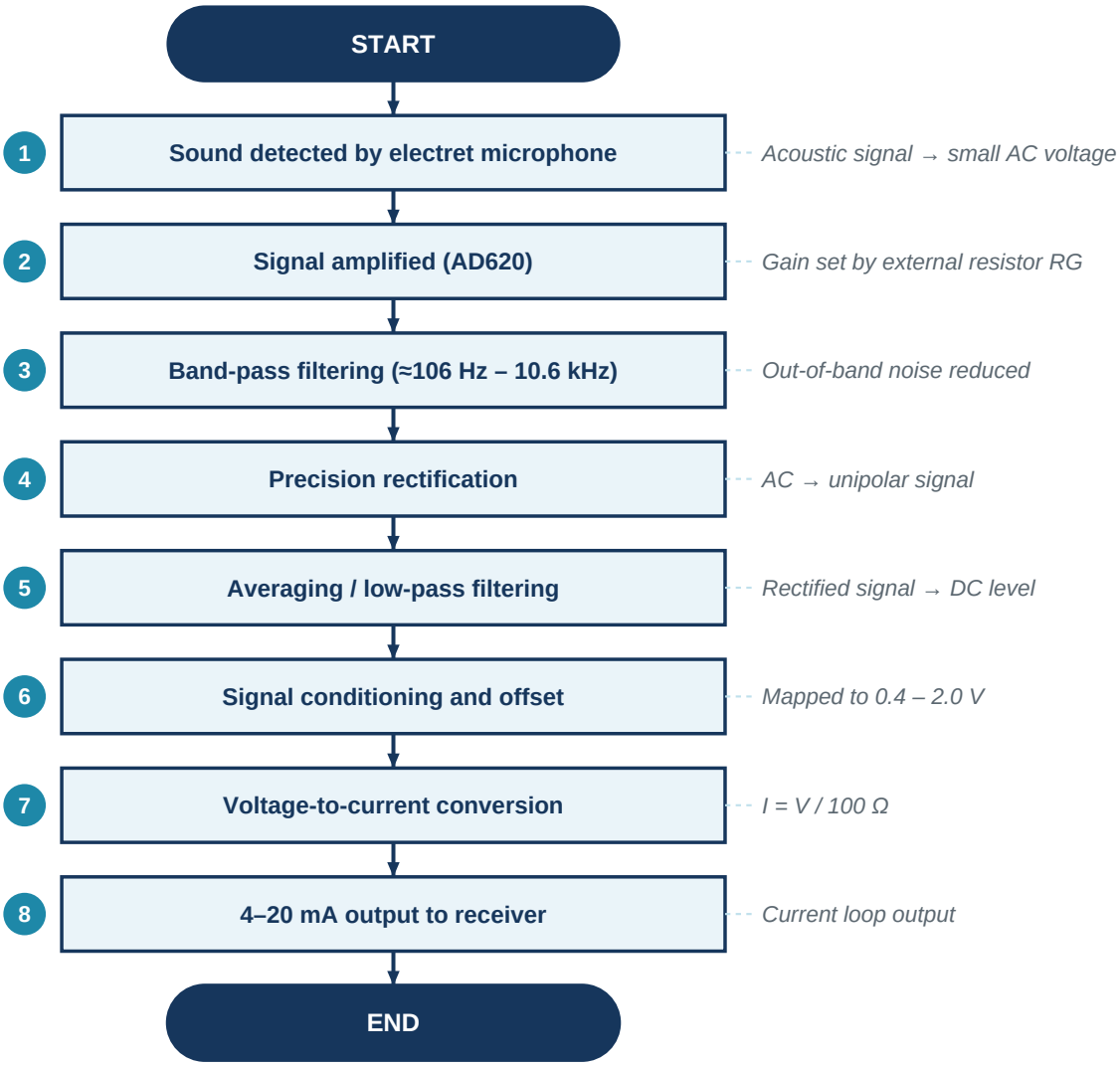


Figure 2: Flowchart of the measurement process.

07 Hardware Design

The circuit was assembled on a breadboard (Figure 3). Its stages and component values follow the LTspice schematic (Figure 5).

Circuit Hardware

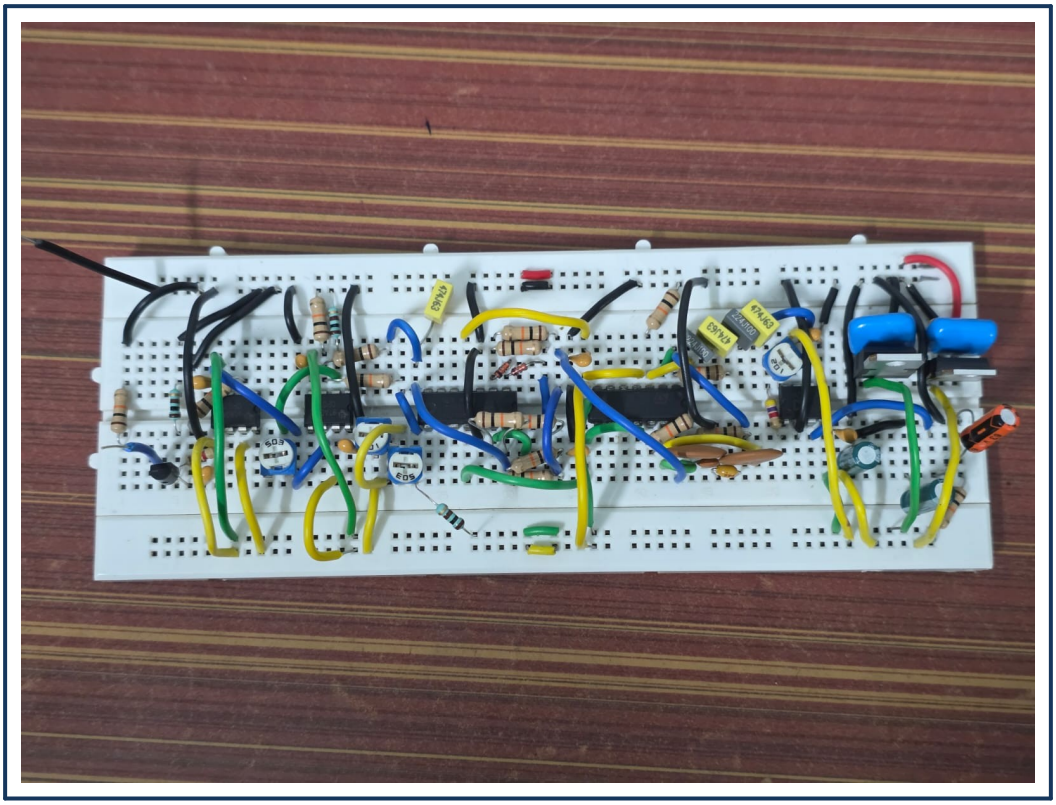


Figure 3(a): Top view of the breadboard circuit.

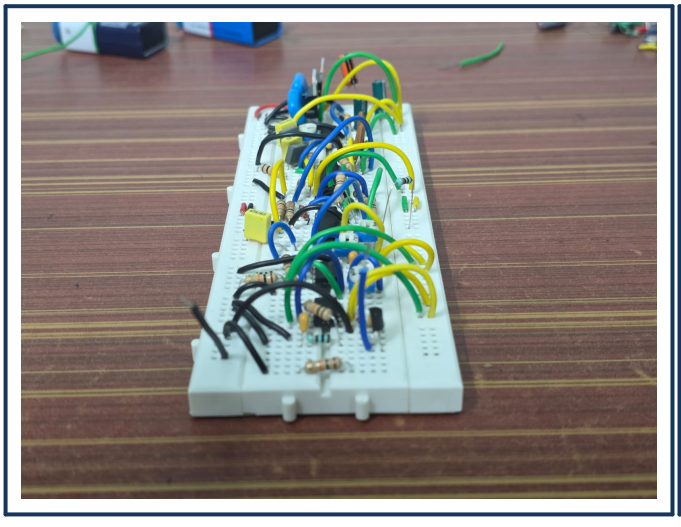


Figure 3(b): Angled view.

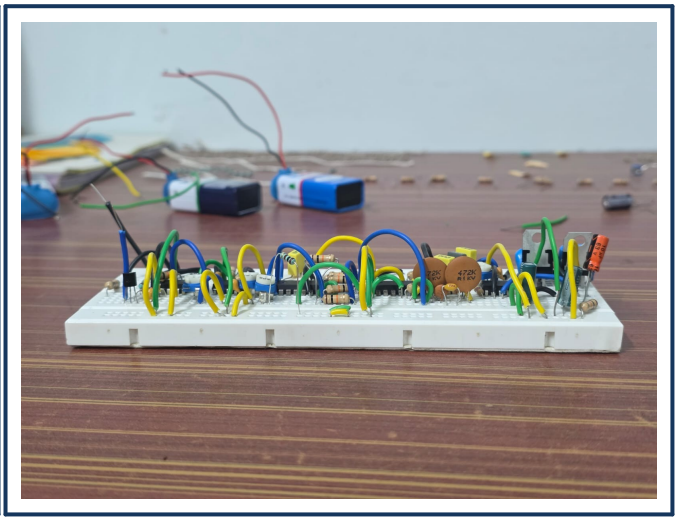


Figure 3(c): Side view.

Circuit Stages and Values

Stage	Main components and values	Function
Input	Electret microphone (simulated by a sine source); C101 = 10 µF	Converts sound to a small AC voltage. Simulation source: 3.98 mV amplitude, 500 Hz.
Amplifier	AD620 (LT1167 model in simulation), RG = 2 kΩ	G = 1 + 49.4 kΩ / RG ≈ 25.7 in the schematic.
High-pass	TL084, 2nd-order Sallen-Key, 2 × 0.15 µF, 2 × 10 kΩ	Natural frequency ≈ 106 Hz; reduces hum and low-frequency noise.
Low-pass	TL084, 2nd-order Sallen-Key, 2 × 1.5 nF, 2 × 10 kΩ	Natural frequency ≈ 10.6 kHz; reduces high-frequency noise.
Precision rectifier	TL084 (2 sections), 2 diodes, 10 kΩ and 5 kΩ resistors	Full-wave rectification of mV-level signals without diode-drop error.
Averaging	500 Ω, 10 kΩ, 10 µF	Smooths the rectified signal into DC; sets the response time.
Gain and offset	OP07, three 10 kΩ inputs, 90.57 kΩ / 1 kΩ feedback, 5.53 mV reference	Maps the DC level to 0.4 – 2.0 V.
Voltage-to-current	OP07, 2N3904, 1 kΩ base, 100 Ω sense, 10 kΩ feedback, 24 V loop	Forces the sense-resistor voltage to equal Vcontrol: I = Vcontrol / 100 Ω.
Power	±12 V (LM7812 / LM7912), 0.1 µF bypass capacitors	Supplies the amplifiers; 24 V supplies the loop.

Circuit Output

Two multimeters were used to read the circuit output during testing (Figure 4). They display 04.6 (left meter) and 1.95 (right meter); the quantity and range of each reading are to be recorded with the measurements.

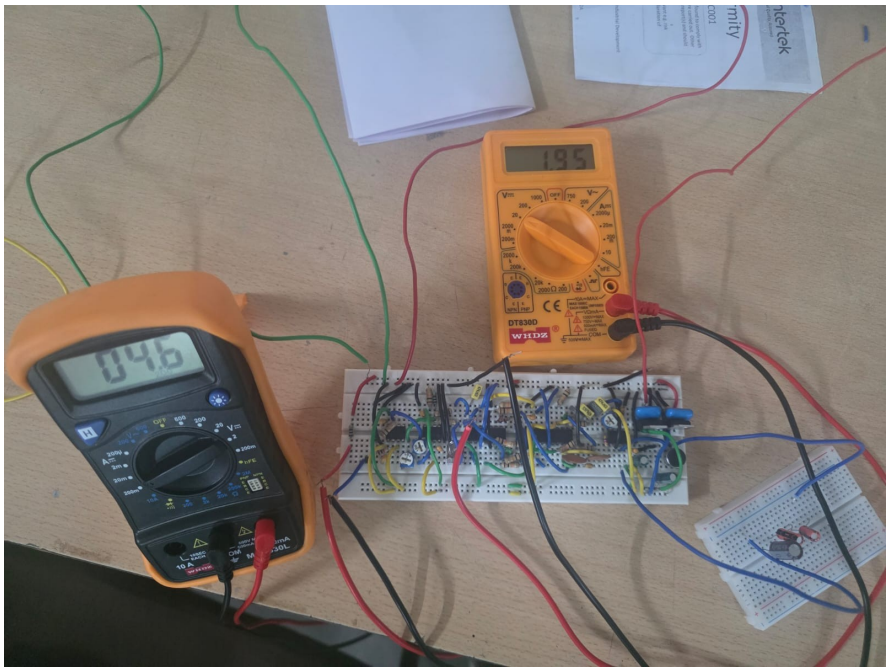


Figure 4: Circuit output on two multimeters.

Design Review and Flagged Items

The supplied design was checked for consistency. Items below should be resolved or verified before the circuit is treated as final.

Priority	Finding	Recommended action
High	**Linear current vs. logarithmic dB.** The detector is amplitude-linear, so 4 mA and 20 mA match 60 dB and 90 dB only at the end points (see Sections 08 and 09).	Add a log/dB conversion stage, or state the output is amplitude-proportional and use a multi-point calibration curve.
High	**Millivolt-level rectifier.** At 60 dB only about 2–5 mV peak reaches the rectifier, comparable to typical TL084 input offset (a few mV) and to diode and bandwidth effects at 10 kHz.	Raise front-end gain and lower later gain; use low-offset op-amps in the rectifier; test linearity at 60 dB.
High	**AD620 input.** C101 is in series with In+ and nothing returns In+ to ground, so the bias current has no DC path. The electret bias network is also not shown (ideal source used).	Add a resistor from In+ to ground and the standard electret bias resistor and supply.
Medium	**Filter response.** Equal-R, equal-C unity-gain Sallen-Key stages have Q = 0.5, not Butterworth (0.707). 106 Hz and 10.6 kHz are natural frequencies; the simulated −3 dB points of the combined response are about 161 Hz and 7.0 kHz.	Change component ratios for Butterworth, or restate the band edges.
Medium	**Values differ** between documentation and schematic: RG 5 kΩ vs 2 kΩ (G ≈ 10.9 vs 25.7); feedback ≈125 kΩ vs 90.57 kΩ; averaging capacitor 100 µF vs 10 µF. The simulation uses 3.98 mV (the 90 dB RMS value) as peak amplitude; its peak is 5.63 mV.	Freeze one set of values and recalculate gain and offset.
Medium	**AD620 mode.** Described as differential, but In− is grounded (single-ended), so cable common-mode rejection is not used.	Use a balanced input if microphone leads are long.
Medium	**2N3904 dissipation.** At 20 mA with 24 V and a 100 Ω load, Vce ≈ 20 V, about 0.4 W, close to the TO-92 limit. Base current adds about 0.5 % error (β ≈ 200).	Use a higher-rated transistor or lower loop voltage; calibrate the final output.
Check	**Rectifier.** The 10 kΩ and 5 kΩ summing resistors match a standard half-wave plus summing circuit, but diode orientation could not be confirmed from the schematic image.	Verify the rectified waveform in simulation and on the bench.
Check	**Loop.** Confirm the 24 V source polarity, the shared ground (no isolation) and the 100 Ω load (receivers are often 250 Ω; this still leaves enough compliance at 24 V).	Verify before connecting a PLC input.
Note	TL084 sections are sufficient (2 filters + 2 rectifier). Near-0 mA is not assumed to indicate a fault; this depends on the receiver.	—

08 Working Principle and Important Calculations

1. The microphone converts sound pressure into a small AC voltage (0.126 – 3.98 mV RMS for 60 – 90 dB).
2. The AD620 amplifies it; the Sallen-Key filters keep roughly 106 Hz – 10.6 kHz.
3. The precision rectifier makes the signal unipolar and the RC network averages it into a DC level.
4. The OP07 stage scales and offsets this level to 0.4 – 2.0 V.
5. The second OP07 drives the 2N3904 so that the voltage across the 100 Ω resistor equals this control voltage, giving 4 – 20 mA in the loop.

Design Values (calculated from the documentation, not measured)

Quantity	60 dB SPL	90 dB SPL
Sound pressure	0.020 Pa	0.632 Pa
Microphone output (RMS)	0.126 mV	3.98 mV
AD620 output, G ≈ 10.9 (RMS)	1.37 mV	43.4 mV
Rectifier peak	1.94 mV	61.4 mV
Average DC level	1.24 mV	39.1 mV
Control voltage	0.4 V	2.0 V
Loop current	4 mA	20 mA

Output at Intermediate Levels (calculated)

Because the detector is amplitude-linear, the current is not linear in dB. The table compares it with an ideal linear-in-dB output between the same end points.

Sound level	Amplitude-linear output (mA)	Ideal linear-in-dB (mA)
60 dB SPL	4.0	4.0
65 dB SPL	4.4	6.7
70 dB SPL	5.1	9.3
75 dB SPL	6.4	12.0
80 dB SPL	8.7	14.7
85 dB SPL	12.8	17.3
90 dB SPL	20.0	20.0

The output can be read as 60–90 dB only after calibration against a reference sound source at several levels, followed by validation with a separate reference meter.

09 Simulation

Circuit Simulation

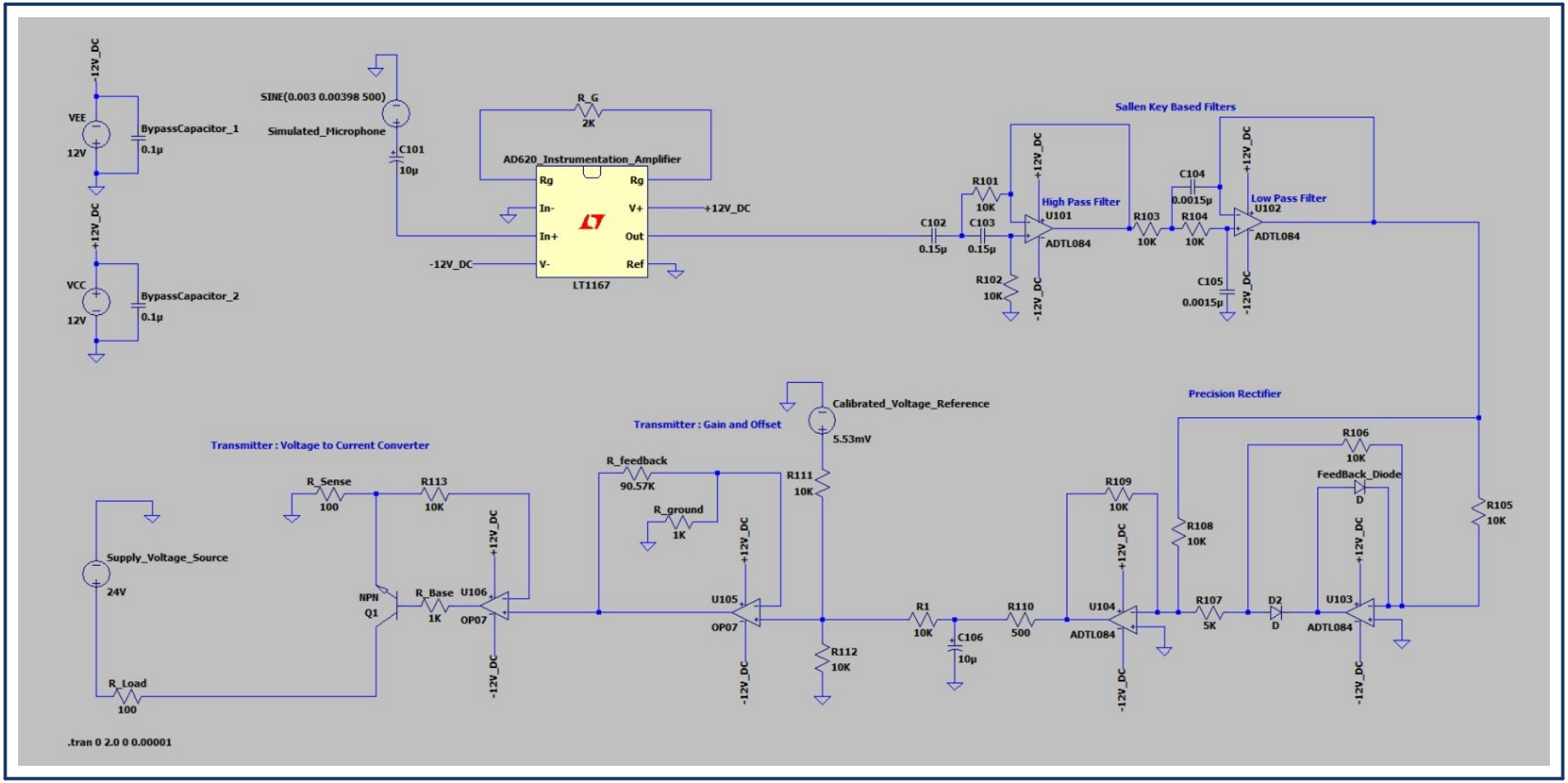


Figure 5: Circuit simulated in LTspice: AD620 front end, Sallen-Key filters, precision rectifier, gain/offset stage and voltage-to-current converter.

Simulation Results

The results below come from a numerical circuit model in Python (SciPy) built from the schematic values: ideal op-amps and diodes, the two Sallen-Key sections as exact second-order transfer functions, the averaging network with the loading of the gain stage, and a 2N3904 with β = 200. The test input is the schematic source (3.98 mV amplitude, 500 Hz; the 3 mV offset is blocked by C101). The level sweep uses a 500 Hz sine whose RMS value follows the −44 dBV/Pa microphone.

Simulated Output at Selected Levels

Sound level	Rectifier average (mV)	Control voltage (V)	Loop current (mA), schematic values	Loop current (mA), re-trimmed
60 dB SPL	2.79	0.25	2.5	4.0
65 dB SPL	4.96	0.32	3.2	4.4
70 dB SPL	8.82	0.43	4.3	5.1
75 dB SPL	15.68	0.63	6.3	6.4
80 dB SPL	27.88	0.99	9.9	8.7
85 dB SPL	49.58	1.64	16.3	12.7
90 dB SPL	88.16	2.78	27.6	19.9

- For the schematic test input the control voltage settles near 2.0 V (about 20 mA) within roughly 30 ms; the response time is set by the averaging network and a small ripple remains at 500 Hz.
- The combined band-pass response has −3 dB points at about 161 Hz and 7.0 kHz, not at 106 Hz and 10.6 kHz.
- With the schematic values the current is 2.5 mA at 60 dB, reaches 4 mA at about 69 dB and 20 mA at about 87 dB, and rises above 20 mA at 90 dB (27.6 mA in the model). The 60–90 dB range is therefore not yet mapped to 4–20 mA.
- Re-trimming the gain stage to about Rf = 57 kΩ and Vref = 17.7 mV gives about 4.0 mA at 60 dB and 19.9 mA at 90 dB in the model. Intermediate levels stay below the ideal linear-in-dB values.

Note: Op-amp offsets, diode dead zone, noise and the microphone bias network are not modelled, so low-level accuracy on the hardware will be worse than shown. These results should be compared with LTspice and bench measurements.

10 Advantages and Applications

Advantages

- Standard 4–20 mA output for PLC/DCS inputs
- Current signal resists cable noise and resistance
- Simple analog design, no microcontroller needed
- Modular stages, easy to test and calibrate
- Averaged output gives a stable reading

Applications

- Noise monitoring near compressors, pumps and fans
- Machinery condition monitoring
- Workplace noise screening (not certified)
- Detecting abnormal events such as leaks
- Trend logging of plant noise in PLC/SCADA

11 Limitations and Future Scope

Limitations

- Calibration against a reference source is required before the output is read as dB.
- Not a standards-compliant sound level meter: no A/C frequency weighting, no specified time weighting, no tested accuracy class.
- Average-responding rather than true RMS, so the reading depends on waveform.
- Output is linear in amplitude, not in dB (see Section 08).
- Low-level accuracy at 60 dB is limited by amplifier offset, noise and microphone sensitivity spread.
- Loop compliance, transistor dissipation and supply stability must be checked.

Future Scope

- Add a logarithmic converter (e.g. AD8307) or a true-RMS converter (e.g. AD736).
- Add A-weighting and fast/slow time response.
- Move to a two-wire loop-powered transmitter (e.g. XTR115) with isolation.
- Build a PCB and add temperature compensation and a local display.

12 Conclusion

The project presents an analog sound level measurement circuit that converts an electret microphone signal into a 4–20 mA current for a 60–90 dB SPL range. A schematic was prepared in LTspice, a breadboard prototype was built, and a circuit-level simulation was run from the schematic values.

The design was reviewed for technical consistency. The main points to resolve are the gain and offset trimming (the simulated span is not yet 60–90 dB), the linear-versus-dB mapping, millivolt-level rectifier accuracy, the AD620 input bias path and mismatched component values. After calibration and validation, the output can serve as an industrial indication of sound level in the 60–90 dB range.

13 References

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